

CYBERSECURITY & ETHICAL HACKING INTERNSHIP

TASK 1 – SECURE NETWORK ASSET INVENTORY

Name: kajal kumari

Project: Secure network asset inventory & system information scanner

Platform: Kali Linux

Repository: secure-network-asset-inventory

1.INTRODUCTION

The Secure Network Asset Inventory project is designed to collect and document basic system and network information from a Linux environment. The project was implemented using Python on Kali Linux. The scanner collects information such as hostname, local IP address, MAC address, operating system details, and available network interfaces. The collected information is saved in a report file for documentation and analysis.

2. OBJECTIVES

To identify basic host information.

To retrieve network details.

To display operating system information.

To generate a structured system and network inventory report.

To save the collected results to a text file.

3. TOOLS & TECHNOLOGIES

Kali Linux – Operating system used for the cybersecurity lab.

Python 3 – Used to develop the network asset and system information scanner.

VirtualBox – Used to run the Kali Linux virtual machine.

Git & GitHub – Used for version control and project repository management.

4. PROJECT DESCRIPTION

Project Title: Secure Network Asset Inventory & System Information Scanner

Problem Statement:

Organizations need an inventory of connected systems to improve asset management and cybersecurity visibility. Manual documentation is slow and often inaccurate. A lightweight Python utility can automatically collect basic system and network information.

Project Description:

The project is a Python-based utility developed in Kali Linux to collect basic system and network information from the host machine. It identifies the hostname, local IP address, MAC address, operating system information, and network interfaces, and generates a structured text report.

5. EXPECTED FEATURES

Hostname Detection – Identifies the hostname of the system.

IP Address Detection – Retrieves the local IP address.

MAC Address Retrieval – Retrieves the MAC address of the system.

Operating System Information – Displays information about the operating system.

Network Interface Listing – Lists the available network interfaces.

Report Generation – Generates and saves the collected information as a text report.

6. IMPLEMENTATION / METHODOLOGY

The project was implemented in the Kali Linux virtual machine using Python. The following steps were followed:

- 1.A Kali Linux environment was configured using VirtualBox.
- 2.A project directory named internship was created.
- 3.The Python program scanner.py was developed.
- 4.The program was executed to collect system and network information.
- 5.The collected information was saved in system_report.txt.
- 6.Basic Linux networking commands such as ip, ping, route, and DNS tools were practiced.
- 7.The project files were organized using Git and pushed to a GitHub repository.

7. PROJECT EXECUTION /RESULTS

The Python-based system information scanner was successfully executed in the Kali Linux environment. The program collected and displayed important system and network information, including the hostname, local IP address, MAC address, operating system information, and available network interfaces. The collected information was also saved in the system_report. txt file for documentation and analysis.

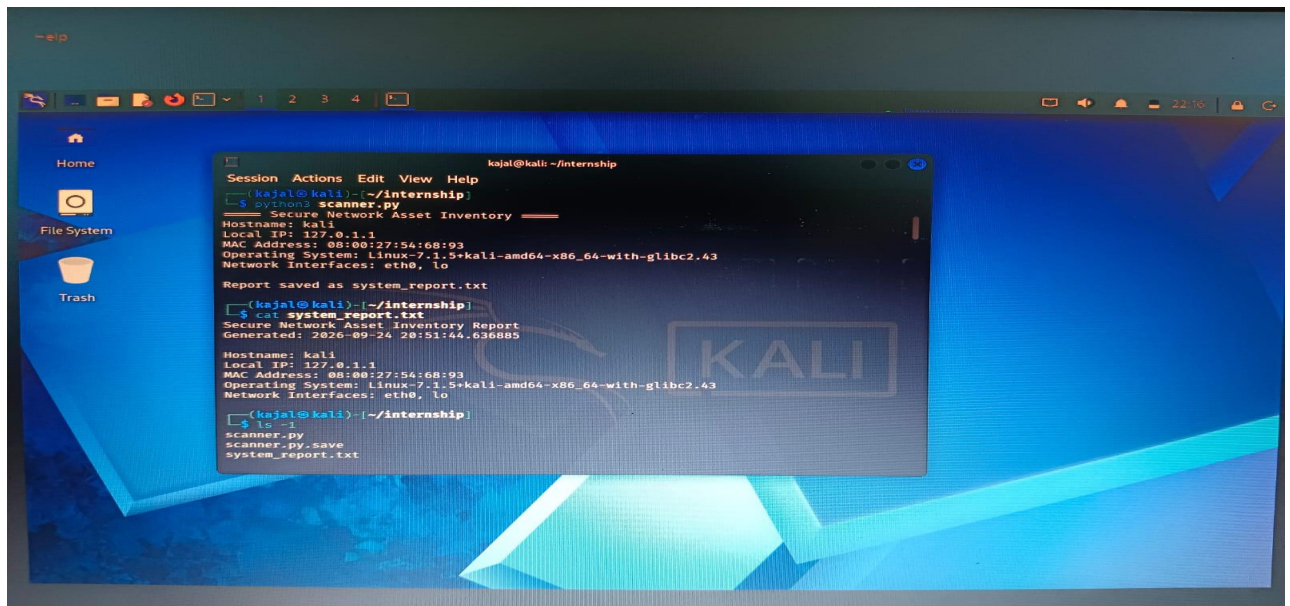


Figure 1: Execution of the Secure Network Asset Inventory scanner in Kali Linux.

8. NETWORKING PRACTICE

Basic Linux networking commands were practiced in the Kali Linux environment to understand network connectivity, routing, and DNS resolution. The ping command was used to test connectivity, ip route was used to view the routing information, and nslookup was used to perform DNS lookup for a domain name. These commands helped verify basic network communication and configuration.

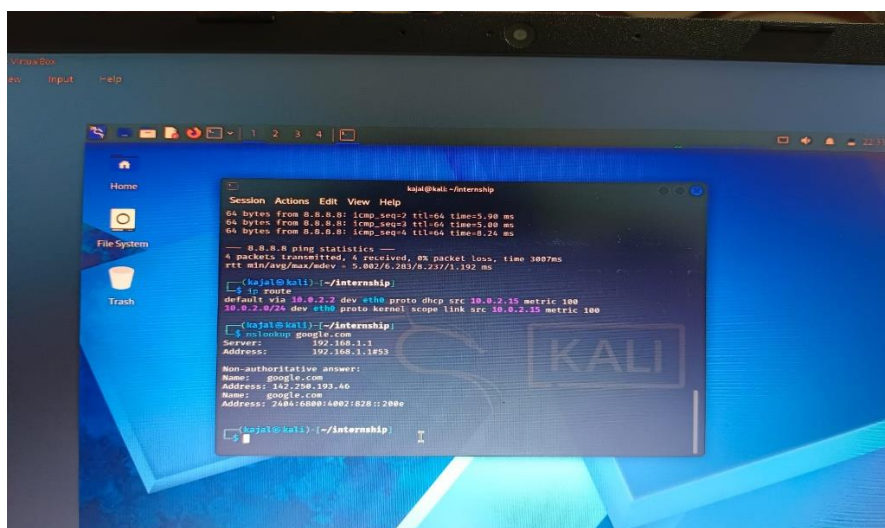


Figure 2: Linux networking commands for connectivity, routing, and DNS testing.

9. GITHUB & PROJECT DOCUMENTATION

The project files were organized using Git and uploaded to a GitHub repository named **secure-network-asset-inventory**. The repository contains the Python scanner,

generated system report, README documentation, and Linux command documentation. GitHub was used to maintain and document the project files.

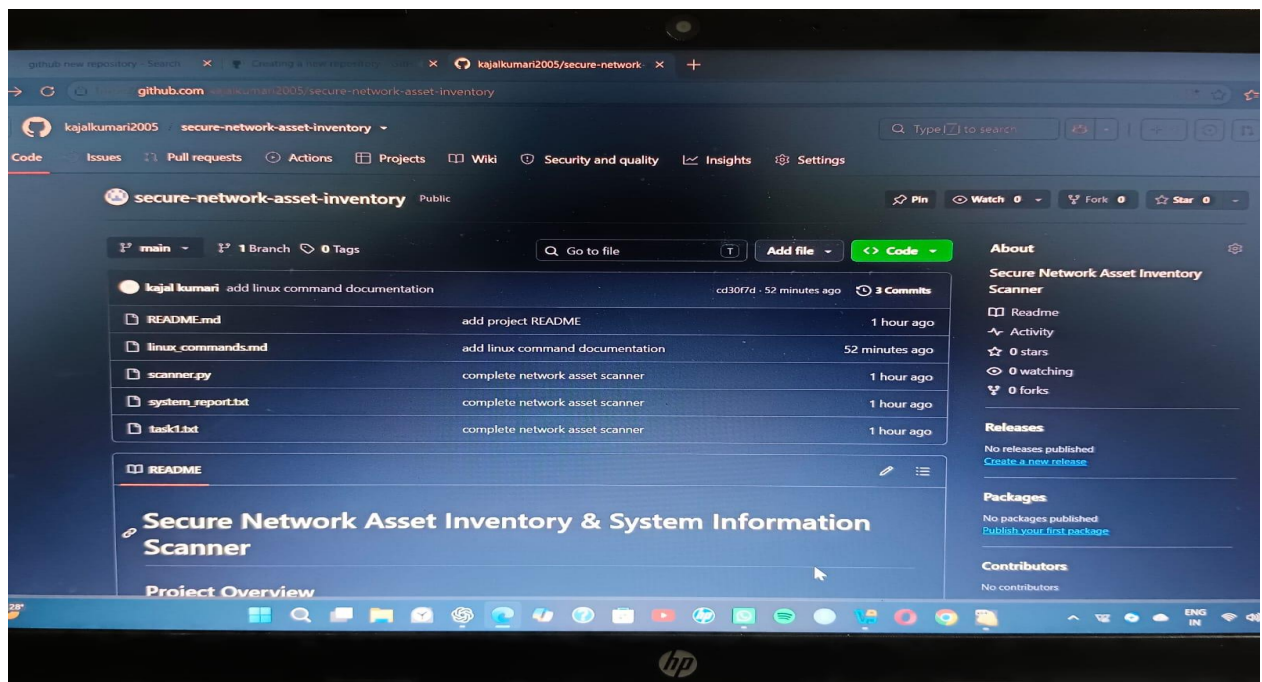


Figure 3: GitHub repository containing the Secure Network Asset Inventory project files.

10. LAB ENVIRONMENT & NETWORK DIAGRAMS

The cybersecurity lab was configured using VirtualBox with Kali Linux as the virtual machine environment. The lab setup provides a controlled environment for practicing Linux commands, networking, Python-based system information collection, and basic cybersecurity activities.

10.1 Cybersecurity Lab Architecture

The lab architecture consists of the host computer, VirtualBox, the Kali Linux virtual machine, and the internal virtual network.

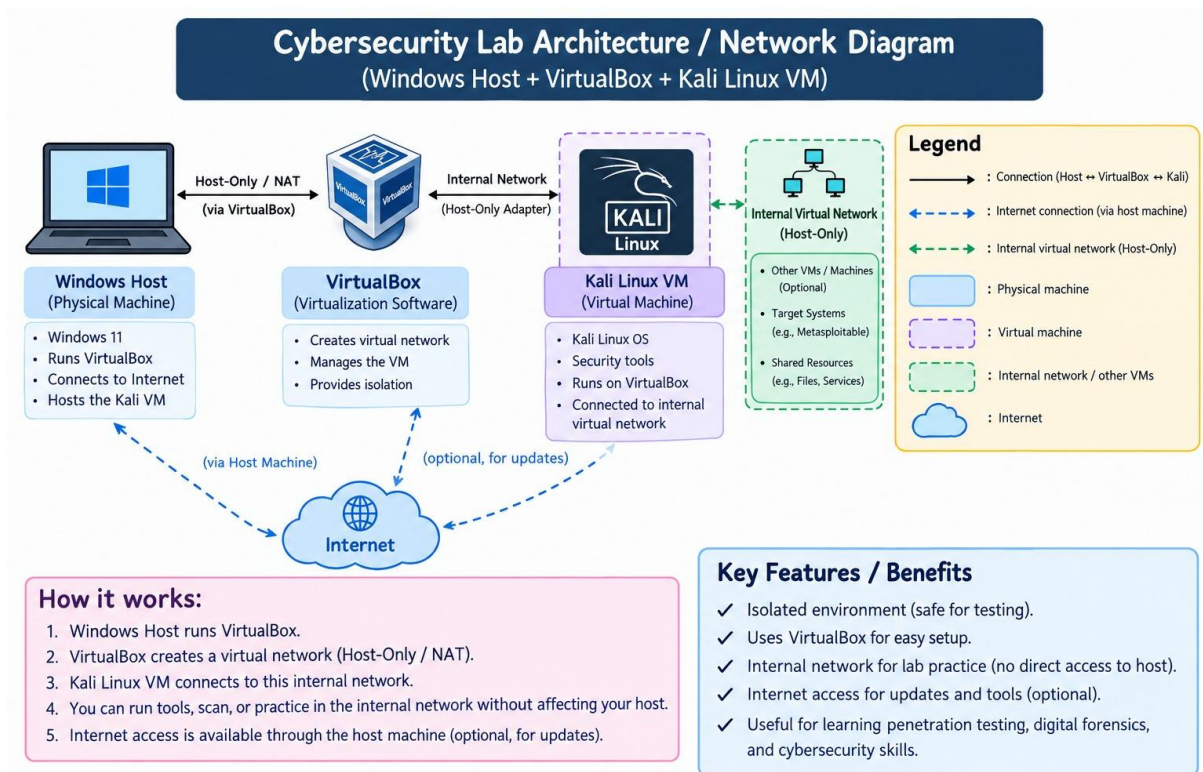


Figure 4: Cybersecurity Lab Architecture.

10.2 OSI Model

The OSI model was studied to understand the seven layers involved in network communication and the role of protocols at different layers.

OSI Model (Open Systems Interconnection)

7 Layers – from Application to Physical

The OSI model is a 7-layer framework that standardizes how data is transmitted across a network, from the application layer (user) to the physical layer (hardware).

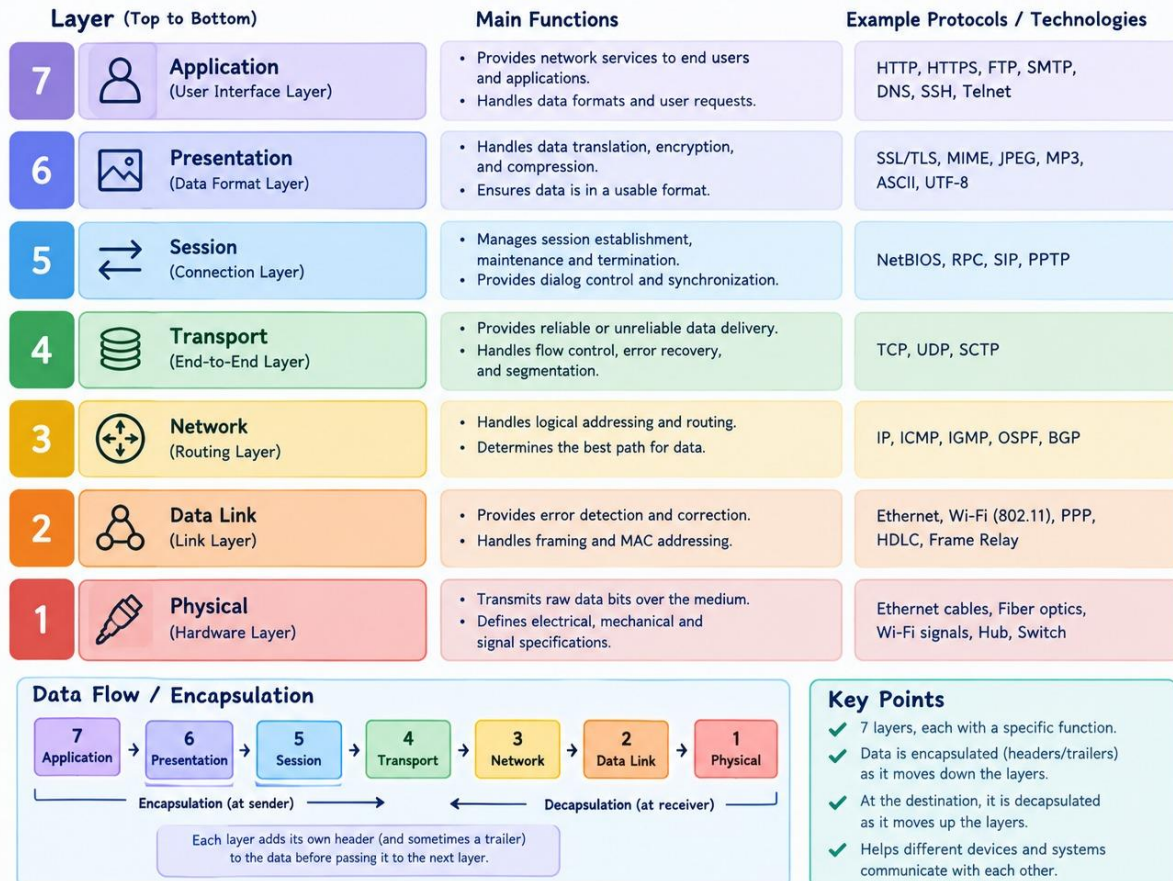


Figure 5: Seven-layer OSI Model.

10.3 TCP/IP Networking

The TCP/IP networking diagram illustrates communication between hosts using IP addressing and common network protocols.

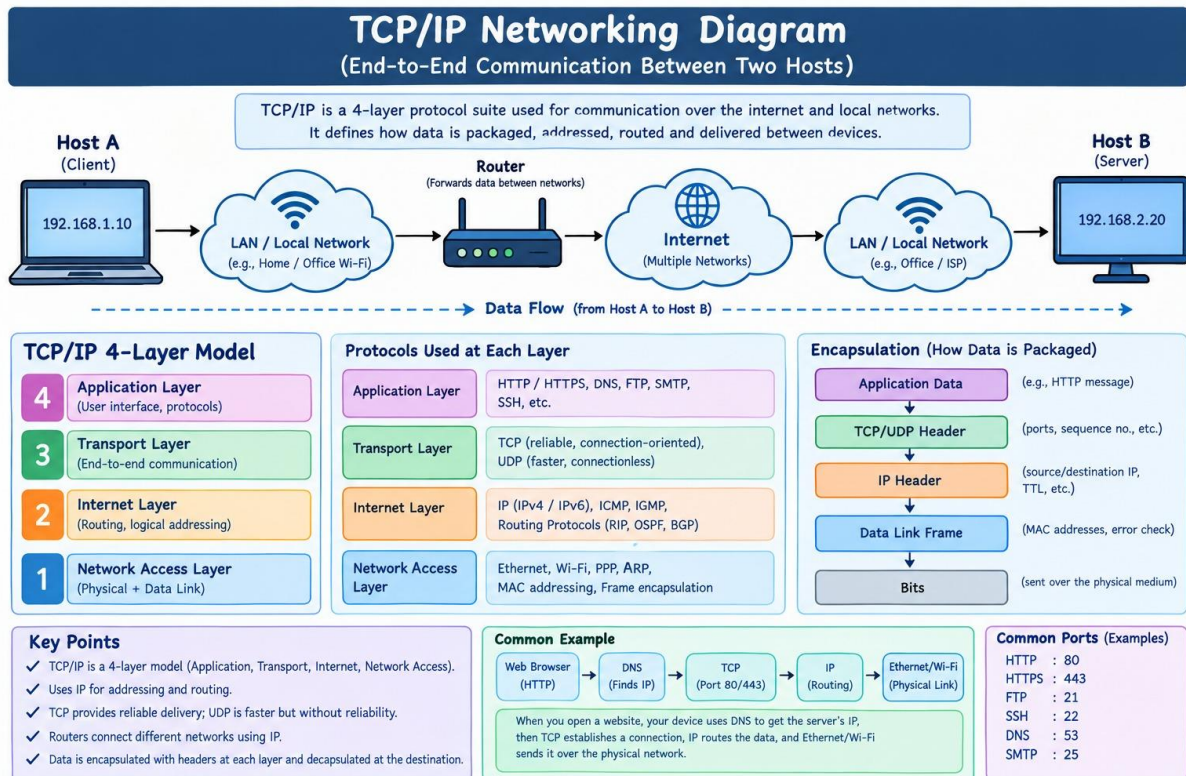


Figure 6: TCP/IP Networking Diagram.

10.4 Linux File System Hierarchy

The Linux file system hierarchy was documented to understand the purpose of important directories such as `/home`, `/etc`, `/var`, `/usr`, and `/bin`.

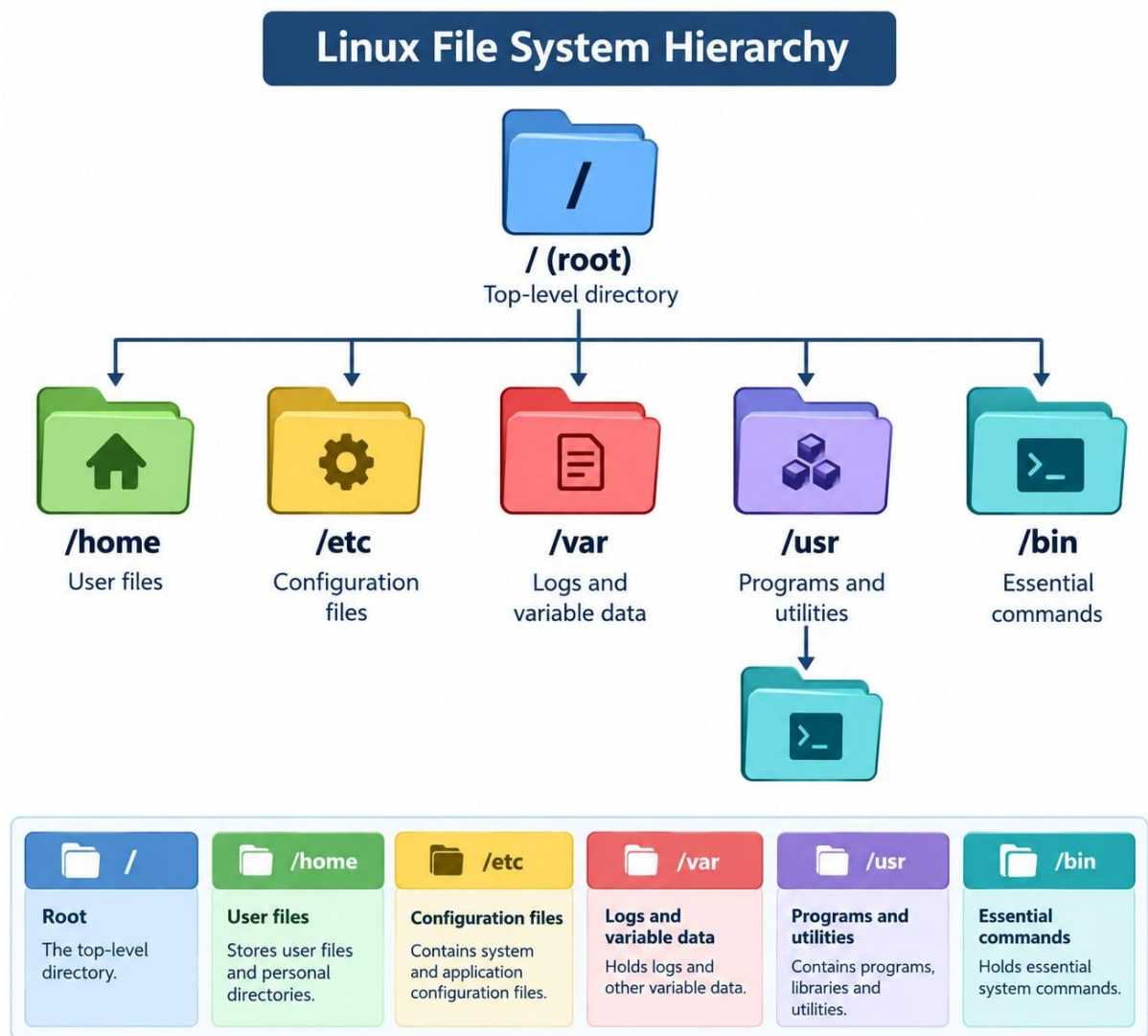


Figure 7: Linux File System Hierarchy.

11. CONCLUSION

The Secure Network Asset Inventory project successfully demonstrated the collection of basic system and network information using Python in a Kali Linux environment. The project covered hostname detection, IP address detection, MAC address retrieval, operating system information, network interface listing, and report generation.

The task also provided practical experience with Kali Linux, VirtualBox, Linux networking commands, Git, and GitHub. The completed project and documentation provide a basic foundation for understanding system information gathering and network asset inventory in a cybersecurity lab environment.

12. PROJECT DELIVERABLES

The completed project includes the required Python source code, system information report, Linux command documentation, GitHub repository, networking and

cybersecurity diagrams, and supporting screenshots. These deliverables document the implementation and practical work completed during Task 1.

Project Files

scanner.py — Python system information scanner

system_report.txt — Generated system information report

README.md — Project documentation

linux_commands.md — Linux command documentation

task1.txt — Task-related project file

GitHub repository — secure-network-asset-inventory

13. REFERENCES

- 1.Cybersecurity & Ethical Hacking Internship – Task 1 guidelines.
- 2.Kali Linux documentation.
- 3.Python documentation.
- 4.Git documentation
- 5.GitHub documentation